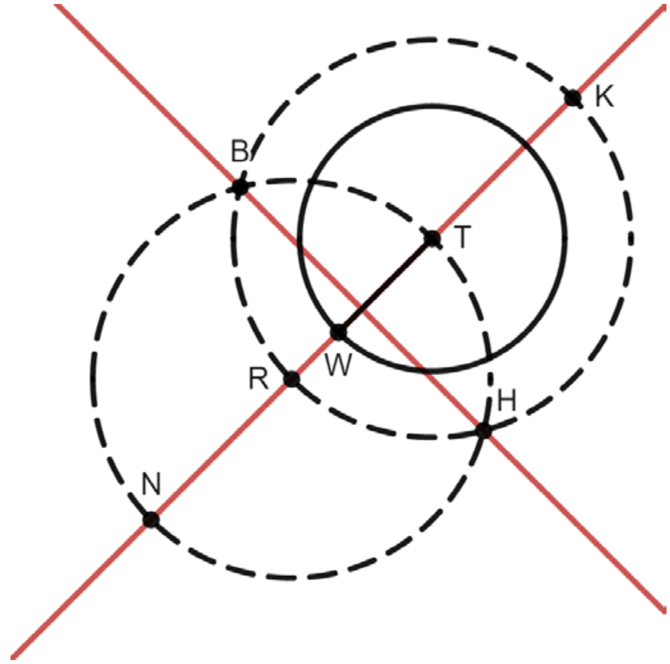
Geometry
Unit 10
Lesson H1 Homework

Name___Answer Key_______
Date____________________
Period___________________

Circle T is shown with radius $\overline{TW}$. The two dotted circles were constructed using a compass so that they are congruent to each other. Points K, H, R, and B are on the dotted Circle T and points B, H, T, and N on the dotted Circle R.



1. Select all of the line segments that are congruent to each other.

- ☐ $\overline{BK}$
- ☐ $\overline{BR}$
- ✖ $\overline{BT}$
- ✖ $\overline{TH}$
- ✖ $\overline{TK}$
- ✖ $\overline{TR}$
- ☐ $\overline{TW}$

2. Justify why the line segments chosen in question 1 are congruent.
Answers may vary. Possible response: Line segments are all radii of Circle T.

3. Select all of the line segments that are congruent to each other.

- ☐ $\overline{BH}$
- ☐ $\overline{BN}$
- ✖ $\overline{BR}$
- ☐ $\overline{NH}$
- ✖ $\overline{NR}$
- ✖ $\overline{RH}$
- ✖ $\overline{RT}$
- ☐ $\overline{RW}$

4. Justify why the line segments chosen in question 3 are congruent.
Answers may vary. Possible response: Line segments are all radii of Circle R.

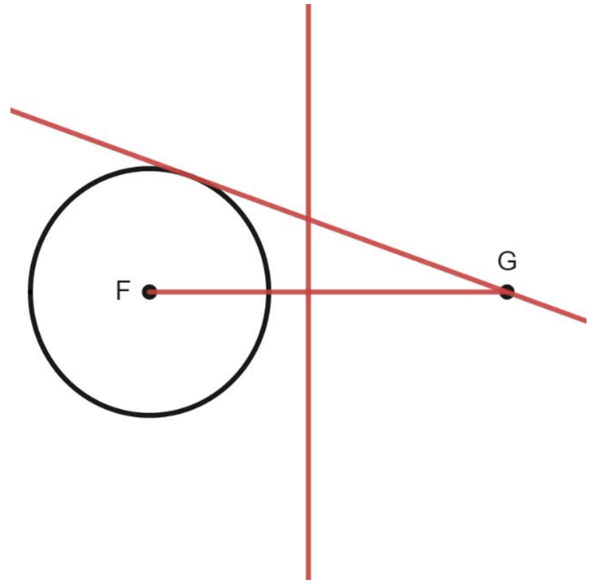
5. Draw $\overleftrightarrow{NK}$ on the diagram.
See diagram.

6. Draw $\overleftrightarrow{BH}$ on the diagram.
See diagram.

7. Describe how $\overleftrightarrow{NK}$ and $\overleftrightarrow{BH}$ are related.
They are perpendicular.

8. Justify whether $\overleftrightarrow{BH}$ is tangent to Circle T.
Answers may vary. Possible response: $\overleftrightarrow{BH}$ is not tangent to Circle T because it intersects the circle twice instead of once.

Circle F and point G are shown.



9. Construct the perpendicular bisector of $\overline{FG}$.
See diagram.

10. Construct a line that is tangent to Circle F and goes through point G.
See diagram.